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Hydrodynamic singularities

When a drop or a bubble pinches off, the flow forgets its past. A pinch of polymer brings the memory back — but only for the drop.

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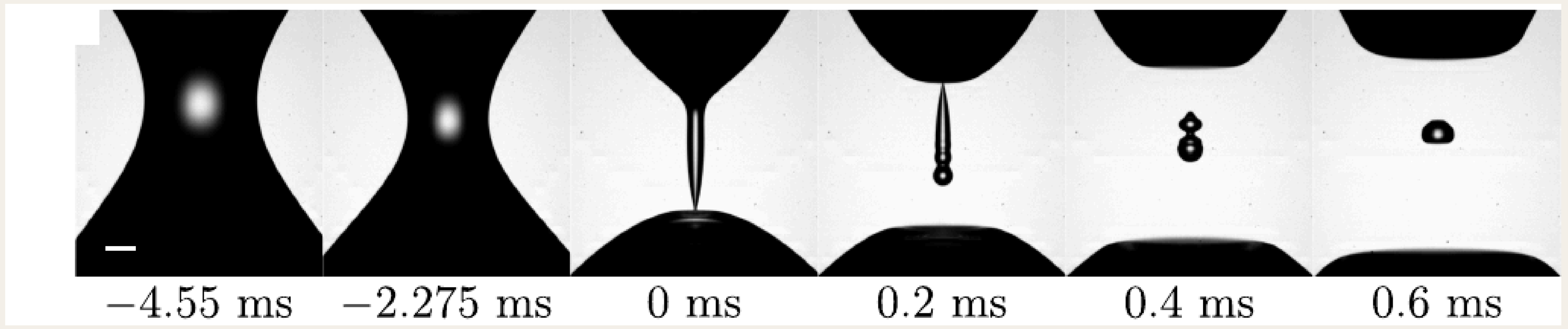
A **singularity** is where a smooth flow runs out of room: a finite quantity racing to infinity in a finite time. As a drop or a bubble pinches off, the neck radius h collapses to zero and the shape turns self-similar, $h \sim (t_0 - t)^{\alpha}$ — a universal form fixed only by the local balance of inertia, surface tension and viscosity. The flow **forgets how it began**. Singularities erase a fluid's memory.

A **drop** pinches off

$h \sim (t_0 - t)^{2/3}$

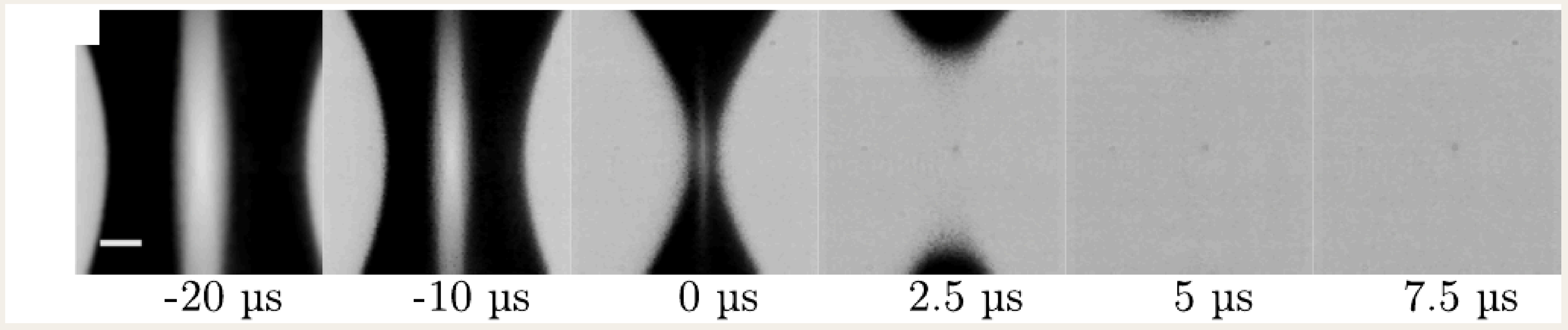
Plain water. Surface tension pulls a slender axial thread to a point, then it snaps into satellite drops — milliseconds.



A **bubble** pinches off

$h \sim (t_0 - t)^{1/2}$

Plain water. The surrounding liquid's inertia collapses the cavity radially — microseconds, a thousand times faster.



Two free surfaces, the same finite-time blow-up: read left to right, $t = 0$ is the singularity. They reach it by different routes (a drop along a thread, a bubble by collapsing a cavity), so they carry different exponents. **What if we gave the fluid a memory?**

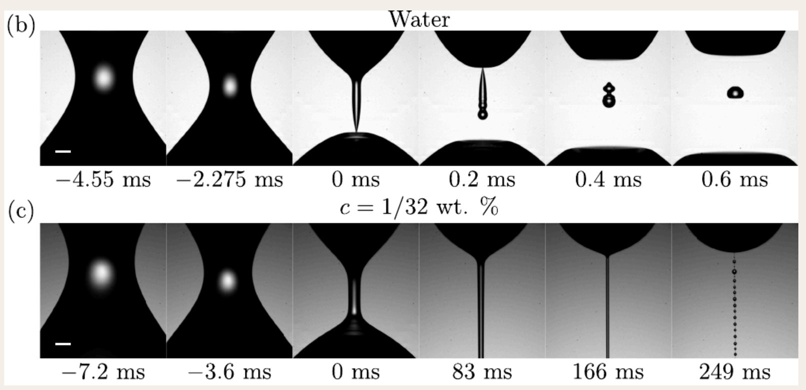
Elasticity is memory

Dissolve a few polymer chains and the liquid gains a memory — each chain stores its own stretching history over a relaxation time. So elasticity should hand back the memory the singularity just erased. It does for a drop, yet a dilute bubble pinches off as if the polymer were not there. Why the difference?

The drop remembers

A water drop reaches a genuine finite-time singularity. The pinch is capillary-driven and the neck is a slender axial thread, so the polymers stretch *along* the axis as it closes.

That alignment makes the elastic stress diverge strongly, $\sigma_{zz} \sim G\,(h_0/h)^4$. It outruns the capillary pull, arrests the pinch and draws the neck into a long-lived **beads-on-a-string** filament — alive for hundreds of milliseconds even at a dilute $c = 1/32$ wt%.



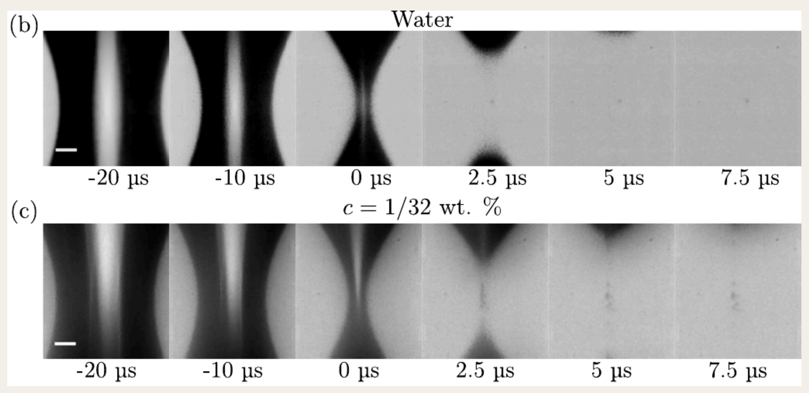
Drop pinch-off. Top: plain water reaches the singularity and snaps into satellites. Bottom: the same dilute polymer draws the neck into a persistent beads-on-a-string thread.

→ Dilute polymer → a thread. Memory wins.

The bubble forgets anyway

A bubble neck pinches just as sharply, $h \sim (t_0 - t)^{1/2}$, but the collapse is inertia-driven and the cavity closes *radially*, so the polymers stretch sideways rather than along a thread.

Stretched the wrong way the elastic stress diverges only weakly, $\sigma_{rr} \sim G\,(h_0/h)^2$, and stays subdominant to the liquid's inertia. A dilute solution cannot organise a filament; the pinch-off stays water-like. A thread needs **high concentration**, and even then its fate hangs on the needle size.



Bubble pinch-off. Top: water. Bottom: the same dilute polymer. In the dilute limit the two near-singular sequences are almost indistinguishable.

→ Dilute polymer → no thread. The singularity survives.

Why elasticity picks sides

	Drop	**Bubble**
WHAT DRIVES THE PINCH	surface tension (capillary)	the outer liquid's inertia
NECK GEOMETRY	a slender **axial** thread — polymers stretch along it	a **radial** cavity — polymers stretch sideways
ELASTIC STRESS	$\sigma_{zz} \sim G(h_0/h)^4$ — strong	$\sigma_{rr} \sim G(h_0/h)^2$ — weak
DOES MEMORY WIN?	yes — it overtakes the drive and grows a thread	no — it stays subdominant; no thread forms

Two extra powers of (h_0/h) is the whole story: along a drop's thread the elastic stress overtakes the drive, while across a bubble's cavity it stays a bystander (the elastocapillary number $\mathrm{Ec} = Gh_0/\gamma$ sets the balance). Reading which singularities a pinch of elasticity can rewrite is how we learn to control breakup — threads, sprays and aerosols.

References

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